

Dashboard Requirements

Functional & Visual Specification — Airline Flight Delay Analysis

Data Source	Maven Analytics — Airline Flight Delays (U.S. DOT, 2015)
Target Tool	Microsoft Power BI
Version	1.1
Status	Final
Date	August 2026

1. Purpose

This document translates the final project goals and KPI definitions into the concrete dashboard specification that was implemented in Power BI. It defines the delivered Executive Overview page, its KPIs, analytical visuals, filters, interactions, and acceptance criteria.

2. Dashboard Structure

The final dashboard consists of **one Executive Overview page**. The page is intentionally focused so that the dashboard remains readable and decision-oriented rather than adding visuals simply to increase density.

#	Page	Primary Purpose
1	Executive Overview	Provide a concise view of 2015 flight volume, delay performance, cancellation patterns, delay severity, airline delay rates, airport delayed-flight concentration, and recorded delay contributors.

3. Dashboard Filters

- **Month Name** — single/selectable month slicer used for temporal filtering.
- **Airline** — airline slicer used to filter the dashboard by carrier.
- No additional slicers are included because the available dashboard space and business requirements do not justify adding filters that do not materially improve decision-making.

4. Executive Overview — KPI Cards

KPI	Purpose
Total Flights	Total scheduled flight records in the selected context.
Delayed Flights	Flights classified as delayed under the project's implemented delay logic.
On-Time Flights	Flights classified as on time under the dashboard status logic.
Delay Rate %	Share of flights classified as delayed under the dashboard measure.
Cancelled Flights	Flights with a cancelled status.
Avg Duration (min)	Average actual flight duration in minutes.

5. Executive Overview — Analytical Visuals

Visual	Type	Purpose / Business Question
Flight Volume & Delays by Month	Combo column + line	How do total flight volume and delayed-flight volume change across months?
Top 5 Airlines by Delay Rate	Horizontal bar	Which airlines show the highest delay rates in the selected context?
Flight Delay Distribution by Duration	Donut	How are delayed flights distributed across duration categories?
Monthly Flight Delay Rate	Line	How does delay rate vary across the year?
Top 5 Origin Airports by Delayed Flights	Horizontal bar	Where is delayed-flight volume concentrated?
Total Delay Minutes by Cause	Horizontal bar	Which recorded delay contributors account for the largest total delay minutes?
Flight Cancellations by Reason	Donut	What recorded reasons account for cancellations?

6. Visual Design & Formatting Requirements

- Use clear, business-readable visual titles that match the final dashboard terminology.
- Use consistent percentage and numeric formatting across KPI cards and analytical visuals.
- Use Top 5 filtering for the airline and origin-airport ranking visuals so scrollbars are not required.
- Maintain readable donut labels and avoid unnecessary visual clutter.
- Keep the executive page compact enough for the KPI row and analytical visuals to remain readable at normal dashboard zoom.
- Do not add decorative visuals that do not answer a documented business question.

7. Interactivity Requirements

- Month and Airline slicers must update the KPI cards and analytical visuals in the expected filter context.
- Standard Power BI cross-filtering/cross-highlighting may be used where it improves analysis.
- Top-5 ranking visuals must retain their Top N behavior when slicers are applied.
- Tooltips should expose exact values where the standard Power BI visual supports them.
- The final Executive Overview does not require a separate reset-filters button or page-navigation structure because it is a single-page deliverable.

8. Non-Functional Requirements

Requirement	Detail
Performance	The dashboard should remain responsive for the approximately 5.8M-row fact table; Import mode and efficient DAX/modeling are preferred.
Readability	Visuals, KPI cards, labels, and titles must remain readable without excessive scrolling or overlapping elements.
Consistency	Percentage KPIs and other metrics must use the definitions documented in the KPI Document and implemented DAX measures consistently.
Usability	The two selected slicers must provide meaningful filtering without overcrowding the executive page.
Documentation	The dashboard and project documentation must clearly distinguish the final implemented Executive Overview from future analytical extensions.

9. Acceptance Criteria

- All six KPI cards defined in Section 4 are present and display the validated project measures.
- All seven analytical visuals defined in Section 5 are present and use the intended fields/measures.
- Month and Airline slicers work correctly and affect the dashboard as expected.
- Top 5 airline and origin-airport visuals display five categories without persistent scrollbars.
- The delay-duration donut and cancellation-reason donut are readable after label cleanup.
- Visual titles and number formats are standardized across the page.
- Dashboard values are consistent with the validated project calculations.
- Every final visual answers a documented business question; no undefined ad-hoc visual is required.
- Findings and recommendations in the final project report are supported by the implemented dashboard.

10. Future Extensions

- Dedicated Airline Performance page with normalized airline benchmarking.
- Dedicated Airport & Route Analysis page with normalized airport metrics and route-level analysis.
- Dedicated Delay Cause Deep-Dive page with controllable versus non-controllable delay analysis.
- Day-of-week and time-of-day analysis.
- Predictive delay forecasting if suitable future data becomes available.

11. Final Implementation Statement

The final Power BI deliverable is the **one-page Executive Overview** described in this document. The future extensions above are retained as possible next phases and are not presented as completed dashboard pages.

Document status: Final — aligned to the delivered Power BI project.